Run AI on someone else’s computer without them seeing your input

Stays on your device	Goes to each worker
Your actual text	A random-looking piece of it
The final answer	The public model + helper numbers

One honest worker is enough

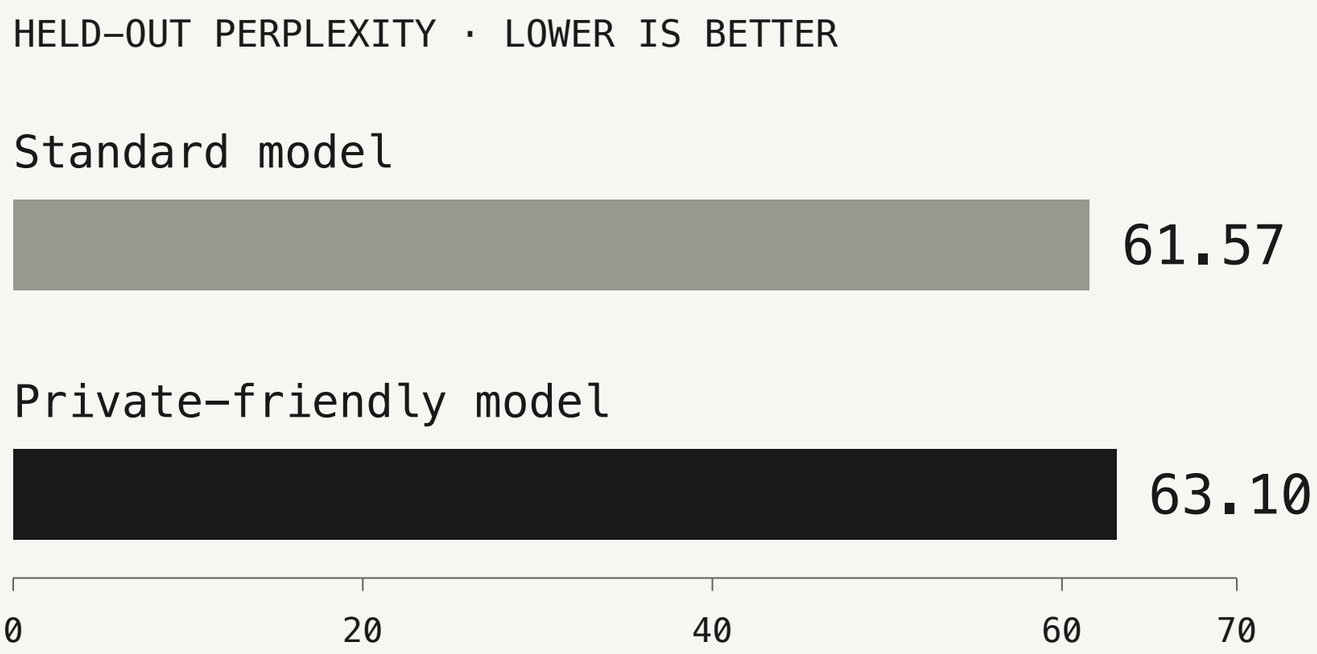
If these workers team up	Can they read your input?
Any one worker	No
Any two workers	No
All three workers	Yes
You run one worker yourself	No

If all three secretly team up, you can’t tell. So pick who runs them.

Assumes the workers are run by different people

In the demo, all three run on our desk

In a separate test, the private-friendly model was 2.48% worse



6,164,096
parameters in each test model

14,336,000
training tokens

Not the demo model.
About quality, not speed.

Measured: perplexity on held-out text · data/quality.json

Separate from the demo

The model only adds, multiplies and squares

Step	Who does it	Messages
Split the input into pieces	Your device	Pieces go to the workers
Multiply by public weights	Each worker, alone	None needed
Square	Workers, via your device	Checked before it’s used
Multiply again, then check	Workers, then your device	Answer pieces come back
Per character		2 round trips

From engine/device.py and engine/node.py

Privacy costs extra compute and messages

What works now, and what’s next

Works now	Next
A 19,295-parameter note model	Real-size language models
Three workers, over the network or USB	Workers run by different people
Same output as the normal model	A full security review
Your device makes the helper numbers	Workers make them and talk directly

Run one worker yourself. Use other people’s computers for the rest.